

SUSHANT LOKHANDE

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SUMMARY

Software Engineer, Machine Learning Systems with experience building **production services, distributed data pipelines, ML inference systems, and reliability tooling**. Strong in **Python, PyTorch, Linux, GPU computing, real-time processing, observability, fault recovery, and visualization**.

TECHNICAL SKILLS

Languages: Python, C++, Rust, TypeScript, Java, SQL, Bash

ML: PyTorch, NumPy, scikit-learn, OpenCV, CNNs, Vision Transformers, Model Inference, Feature Extraction

Systems: Linux, Distributed Systems, Multithreading, Async Programming, GPU Computing, CUDA, Docker

Data: Real-Time Pipelines, Stream Processing, PostgreSQL, Redis, Kafka, Missing/Late Data Handling

Reliability: Monitoring, Structured Logging, Metrics, Tracing, Fault Recovery, Retries, Health Checks, Automated Testing

Engineering: FastAPI, React, TypeScript, REST APIs, Git, CI/CD, Profiling, Visualization, Claude Code, Cursor, Codex

EXPERIENCE

San Jose State University | *Software Engineer, Machine Learning Systems*

Aug 2025 – May 2026

- **Reduced distributed experiment setup effort by 55%** by building Python tooling for workload orchestration, configuration, telemetry capture, failure detection, replay, and automated analysis across **1,200+ runs**.
- **Improved streaming reliability by 19%** versus a fixed-rate baseline by processing real-time network observations and adapting execution under changing bandwidth, latency, packet loss, and failure conditions.
- **Built GPU-accelerated PyTorch inference pipelines across 136K derived images** for master's thesis research, combining CNN, ResNet50, VGG16, ViT-B/16, HOG, and Haralick representations for feature extraction and classification.
- **Improved classification accuracy to 80.2% across 17 classes** by combining complementary handcrafted and deep image features through soft-voting ensembles, outperforming the strongest standalone model by **2.4 percentage points**.
- **Expanded failure analysis across 14 interoperability scenarios** by converting recurring system, network, and dependency failures into repeatable tests with structured telemetry and progress visibility.

Turito | *Software Engineer, Production Systems*

Jun 2022 – Oct 2022

- **Cut controlled-disruption failures by 42%, from 3.1% to 1.8%** by hardening retries, timeout handling, recovery logic, and distributed state transitions in always-on real-time services.
- **Slashed p95 CPU utilization by 22%** by profiling high-frequency processing paths and eliminating excess allocations, payload copies, routing lookups, and synchronization overhead.
- **Lowered network egress by 14%** by simplifying data-routing and buffering paths while improving reuse across high-frequency streaming workloads.
- **Reduced production diagnosis time by 35%** by adding structured metrics, runtime-state visibility, targeted logs, health signals, and failure dashboards for production services.

Trivia Softwares | *Software Engineering Intern*

Nov 2021 – Jan 2022

- **Scaled an asynchronous processing framework to 25,000 logical sessions** using Rust/Tokio workers and configurable streaming workloads to reproduce concurrency and cleanup failures.
- **Improved equivalent-load throughput by 31%** by identifying lock contention, serialized writes, and slow cleanup using profiling and stress testing.
- **Reduced investigation effort by 40%** by building Python aggregation, SQLite-backed analysis, and Flask/Chart.js visualization tooling for latency, throughput, runtime state, and failures.

RESEARCH & PROJECTS

Image-Based Techniques and Ensemble Soft Voting for Malware Classification | *PyTorch, Computer Vision, GPU*

- **Processed 17K malware samples into approximately 136K images across 8 representations**, extracting visual patterns through HOG, Haralick, pretrained CNN/transformer embeddings, and a custom 512-dimensional CNN representation.
- **Reached 80.2% accuracy across 17 malware families** by combining 15 complementary classifiers through soft voting.

Forecast Ops | *Python, PyTorch, Kafka, PostgreSQL, React, Docker, GPU*

- **Maintained continuous model inference during simulated node failures** by adding heartbeat detection, checkpointed jobs, automatic retries, workload rescheduling, and idempotent processing across distributed workers.
- **Processed 2.4M time-stamped observations** through a streaming pipeline that handled late, duplicated, and missing records using event-time windows, watermarks, validation rules, and replayable ingestion.
- **Cut failure localization time by 62%** with a React monitoring dashboard in real time.

EDUCATION

San Jose State University | M.S. Computer Science | GPA: 3.79/4.0

Aug 2024 – May 2026